

RU Parking?

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57%

Commuters

Audience: Rutgers University-New Brunswick

Primary Stakeholder: College Avenue
Commuter Students (with cars)

Secondary Stakeholders: Rutgers
Transportation & Parking Services



r/rutgers • 5mo ago

CabbageSass

College Ave lots full

Anyone else have a hard time finding a spot? Never had a problem until this semester and I do have a pass. Anyone else notice it's way more crowded this semester? Gonna have to get there at 7 am for my 8:30 class I guess.

16

 12

r/rutgers • 3y ago

Deshes011

Bruh that's not a parking spot

r/rutgers • 4y ago

Tall-Soy-Latte

Parking has gotten so bad people are just making up spots now lol

104

 9

Solution Overview



CCTV
cameras+Sensors

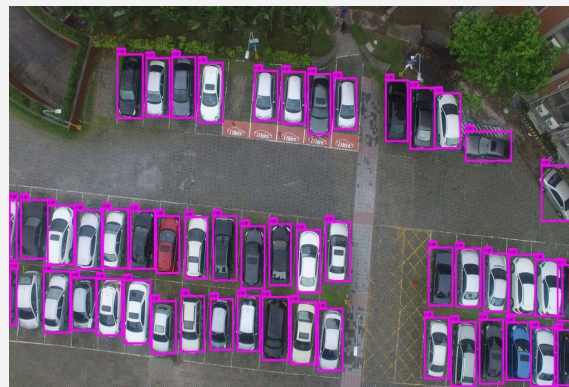
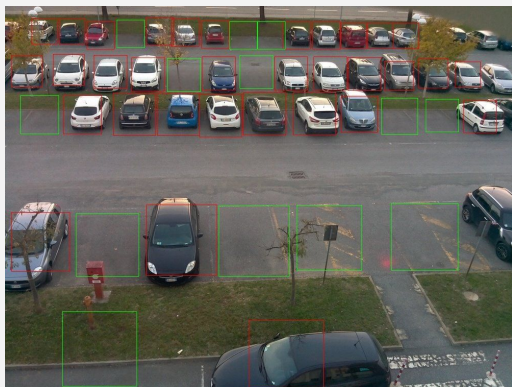
Detection Using
Deep learning
models

Data
integration/Mobile
Application for
student use



Deep Learning Models

- Yolo: You only look once
- Looks at entire image in one pass
Fast 30-60 fps
- Accuracy: approximately 95%
- R-CNN(Region Based CNN)
- Checks regions carefully one by one, slow 1-2 sec per image
- Accuracy: Approximately 98%

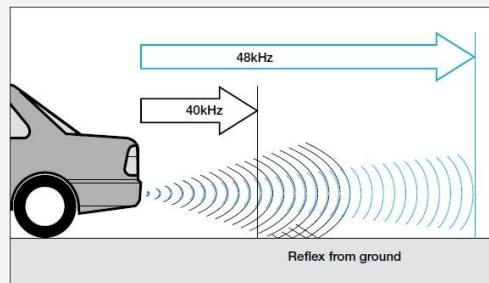


<https://www.inovex.de/de/blog/machine-learning-on-the-edge-parking-guidance-systems/>

Sensors



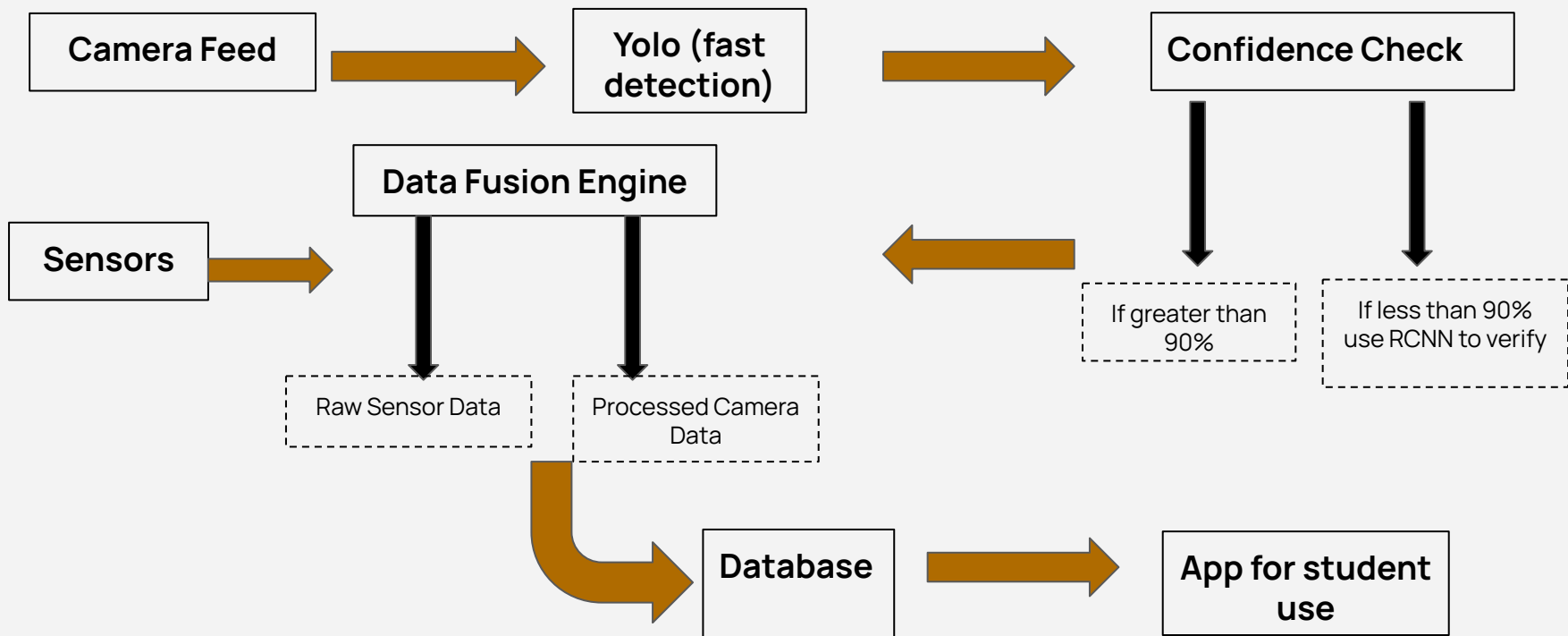
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<https://www.newelectronics.co.uk/content/features/an-introduction-to-ultrasonic-sensors-for-vehicle-parking>

Sensor Type	How it Works/ Placement	Weather/Best Use
Magnetometer	Detects metal via magnetic field / Buried in-ground	Excellent in all weather / Outdoor parking lots
Infrared	Sound waves measure echo / Mounted on ceiling/pole	Good, but affected by heavy rain/snow / Garages, covered lots
Ultrasonic	Beam detects interruption / Mounted on pole/wall	Fair, affected by fog/rain/sun glare / Indoor only

Data Processing





Why This Framework?

- Traditional software fails with different lighting, weather, and camera angles. Deep learning detects cars reliably in rain, fog, sun, shade, and darkness.
- Real-Time Speed + High Accuracy
- YOLO processes 30-60 frames per second → live updates
- R-CNN verifies tricky spots with ~98% accuracy
- Best of both: fast when possible, accurate when needed; this also ensures that we would not have to buy as many sensors



Benefits of Framework

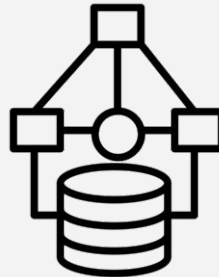
- Provides real-time parking availability in lots and decks before students arrive.
- Improves commute planning by helping students choose the best lot in advance.
- Reduces lateness by minimizing time spent searching for parking.
- Balances accuracy and cost through scalable and efficient technology.
- Prevents overcrowding by redirecting drivers to available lots early.

Limitations

Weather



Model Accuracy Issues



Maintenance



Financial Breakdown

Sensors

Most Expensive

CCTVs

Relatively Expensive

App Development Cost

One time major cost, with frequent update costs

Maintenance

Accumulates expenses over time, but annually being the least expensive

Financial Breakdown

Item	Estimates	Notes
One-time training	\$500 - \$2,000	Fine-tuning pre-trained models
Monthly Inference	\$300 - \$800	Running Yolo on Camera Feed
Monthly Storage	\$50 - \$150	Footage + Database
Year 1 Total	~\$5,000 - \$12,000	Training + 12 months operation

***These estimates are based on AWS pricing and best practices for machine learning optimization.**

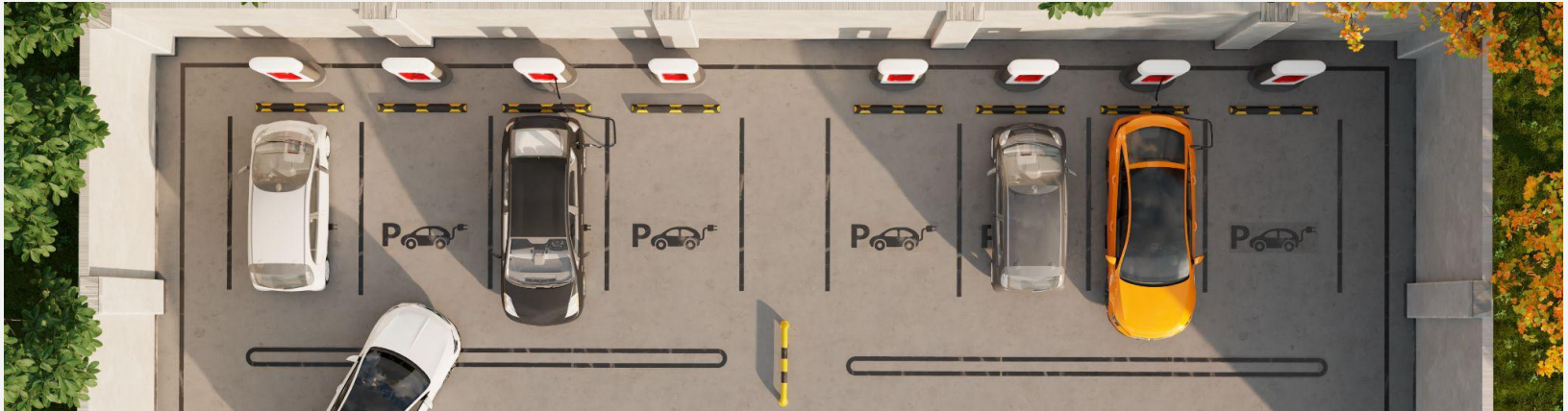
Implications

Negative

- Poor performance in the application will result in lower user traffic
- Privacy concerns with CCTV and sensors

Positive

- Reduction in traffic congestion
- Minimizes unnecessary arrival time buffers



Next Step



Collaboration

Collaborating with Commuter
representatives of RUSA (Rutgers
University Student Assembly) &
Rutgers Commuter Student
Association



Map Feature

Color-coded map
Green = Available
Yellow = Limited
Red = Full

Alternative Parking Spots

Refer to secondary or overflow
parking areas suggested to users
when their preferred parking lot is
full or nearing capacity.



Resources

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